

Correlation & Regression

Consolidation

$$1) S_{xx} = \sum x^2 - \frac{(\sum x)^2}{20}$$

$$= 235724 - \frac{2192^2}{20}$$

$$= 41668$$

$$S_{xy} = \sum xy - \frac{\sum x \sum y}{20}$$

$$= 24758$$

$$S_{yy} = \sum y^2 - \frac{(\sum y)^2}{20}$$

$$= 7818.55$$

$$r = \frac{24758}{\sqrt{41668 \times 7818.55}} = 0.4248$$

$$= 0.425$$

$$H_0: r = 0$$

$$H_1: r > 0$$

At 5% significance, crit. value = 0.3763 for $n=20$

$|r| > 0.3763$, so sufficient evidence to reject H_0 that $r = 0$.

ii) If significance level is different, H_0 may not be rejected.

iii) The significance level is the probability that H_0 will be rejected if it is true

1 2 3 4 5 6 7 8 9 10
B, A, F, G, C, I, E, J, H, D

2 10 1 1 2 6 2 3 1 3 2
J 1 1 4 3 4 4 1 4 4

$$\text{Spearman rank} = 1 - \frac{6 \sum d^2}{n(n^2-1)}$$

$$= 1 - \frac{6 \times 68}{10 \times 99}$$

$$r_s = 0.588$$

$$H_0: r_s = 0$$

$$H_1: r_s > 0$$

At 5% significance, with $n=10$:

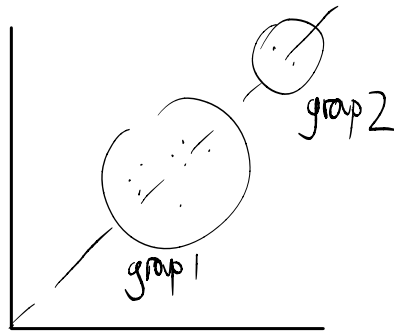
critical value $r_s = 0.5636$

$$0.588 > 0.5636$$

So sufficient evidence to reject H_0 that there is no association between performance on floor exercises & parallel bars

iii) the p.m.c.c. is higher because the two groups of data appear to be positively correlated

However, the two high scores may be outliers, so the Spearman's rank correlation coefficient is better suited because outliers are present.



$$3 \quad \rho = r = \frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}}$$

$$S_{xx} = 3853 - \frac{207^2}{25} = 83.04$$

$$S_{yy} = 3008 - \frac{250^2}{25} = 608$$

$$S_{xy} = 3143 - \frac{207 \times 250}{25} = 73$$

$$\therefore \rho = \frac{73}{\sqrt{83.04 \times 608}} = 0.355$$

$$H_0: \rho = 0$$

$$H_1: \rho \neq 0$$

At $n=25$, at 5% correlation (2-tailed test):
critical value = 0.3961.

$$0.3554 < 0.3961$$

So insufficient evidence to reject H_0 that there is correlation between temperature and wind speed.

ii) The distribution has to be Normal bivariate.
scatter diagram has elliptical shape. $\therefore$ reasonable assumption

iii) It would cause the p.m.c.c. to change substantially so the test would be less valid.
Spearman's rank test would be better.

Exam Questions

a)

Judge 1	rank	7	6	5	4	3	2	1
Judge 2		6	4	7	5	3	2	1
d_j		1	2	2	1	0	0	0
d_j^2		1	4	4	1	0	0	0

Spearman's rank coefficient $r_s = 1 - \frac{6 \times 10}{7 \times 48}$
 $= 0.8214$

$$H_0: r_s = 0$$

$$H_1: r_s \neq 0$$

For 7 values, critical value @ 1% significance = 0.4286

$$0.8214 < 0.4286$$

$\therefore$ Not enough evidence to reject H_0 that there is not a high level of agreement between the two judges.

a) $r_s = 1 - \frac{6 \times \sum d_j^2}{7} = 1$

$$\therefore \frac{6 \times \sum d_j^2}{7} = 0$$

$$\sum d_j^2 = 0$$

$\therefore$ The competitor must randomly pick the order 7654321

$$\text{so } \frac{1}{7!} = \text{60th chance}$$

$$= 0.0168\% \text{ chance}$$

i) $r_s = 1 - \frac{6 \times \sum d_j^2}{7} = \frac{85}{86}$

$$\therefore \frac{6 \times \sum d_j^2}{7} = \frac{1}{86}$$

$$6 \times \sum d_j^2 = \frac{1}{8}$$

$$\sum d_j^2 = \frac{3}{4}$$

but d is integer $\therefore \sum d_j^2$ is integer

so cannot be $\frac{3}{4} = 0.75$ ($\nexists \in \mathbb{Z}$)

$$2a) S_{xx} = 227.45 - \frac{32.1^2}{5} = 8.828$$

$$S_{yy} = 664.26 - \frac{56.6^2}{5} = 23.548$$

$$S_{xy} = 362.37 - \frac{32.1 \times 56.6}{5} = -12.322$$

$$r = \frac{-12.322}{\sqrt{8.828 \times 23.548}} \\ = -0.8546$$

b) the data may not be nonlinear, or there are outliers.
This causes the data to not display a strong trend.

$$c) H_0: r = 0 \\ H_1: r \neq 0$$

critical value for 2-tailed test @ 5% significance
= 0.8783

$$|r| = 0.8546$$

$$0.8546 < 0.8783$$

So insufficient evidence to reject H_0 that there is correlation between x and y

$$d) \begin{array}{c|ccccc} \text{rank}_x & 1 & 2 & 3 & 4 & 5 \\ \text{rank}_y & 5 & 4 & 3 & 2 & 1 \\ & 4 & 2 & 0 & 2 & 4 \end{array}$$

$$16 \quad 4 \quad 0 \quad 4 \quad 16$$

$$\sum d^2 = 40$$

$$\therefore r_s = 1 - \frac{10 \times 6}{5 \times 24} \\ = 1 - 2 \\ = -1$$

e) i) r : does not change
 r_s : does not change

ii) r : changes
 r_s : does not change